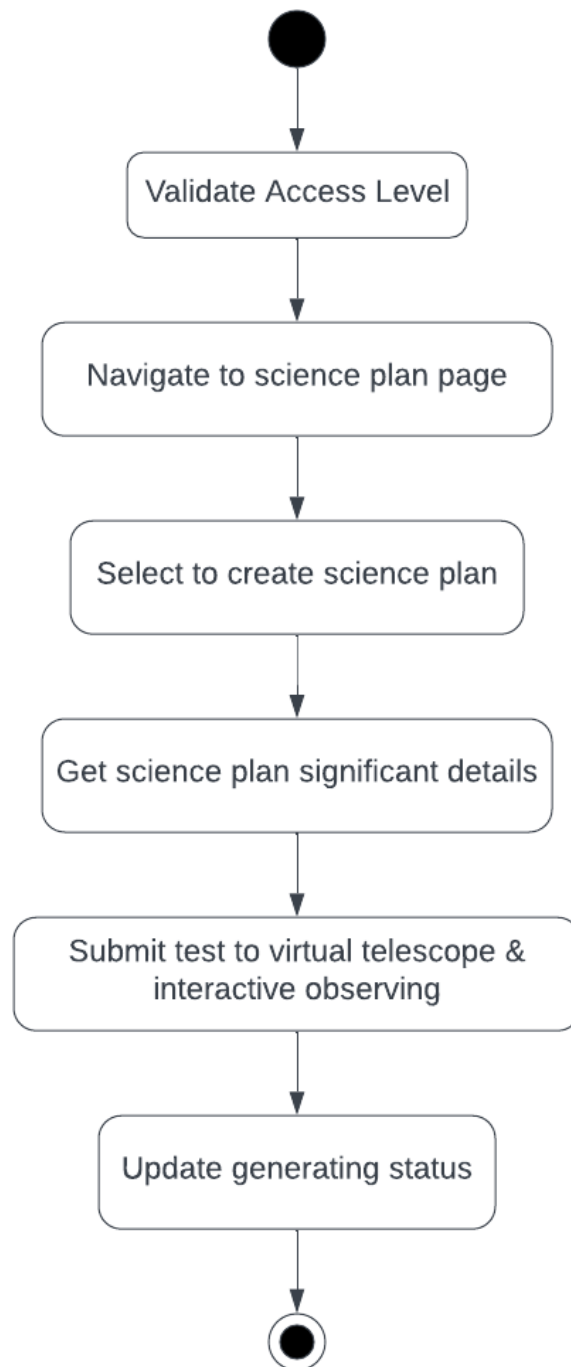
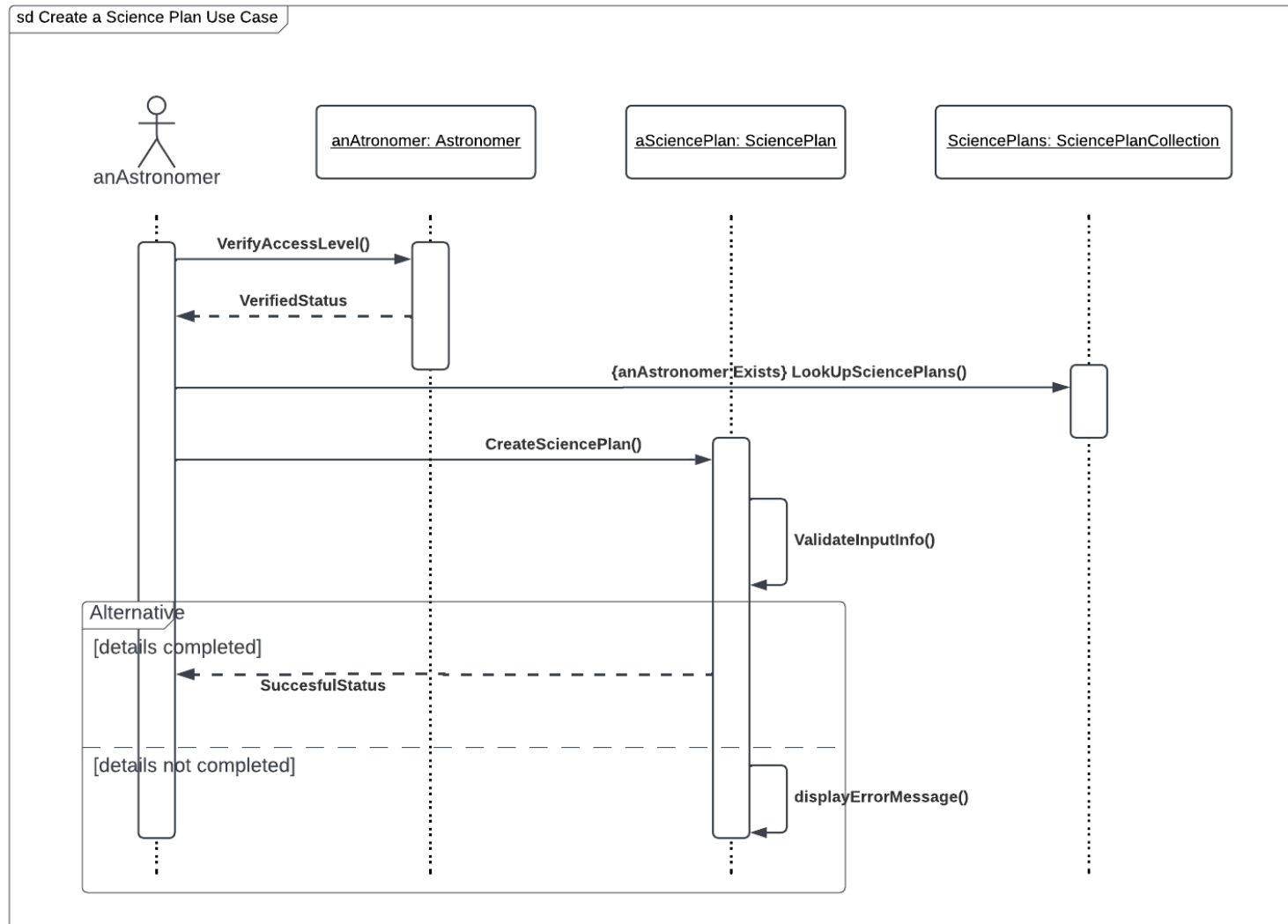


Use Case Name: Create a science plan	ID: 1	Importance Level: High
Primary Actor: Astronomer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science Observer		
Brief Description: This use case describes how science is created.		
Trigger: Astronomers need to propose a new science plan for collecting astronomical data.		
Type: External		
Relationships: Association: Astronomer Include: Extend: Generalization:		
Normal Flow of Events: 1. The system validates the user's level of access. 2. The system navigates the user to the science plan page. 3. The astronomer selects to create a science plan. 4. The system provides a template that the astronomer can fill in. 5. The astronomer fills in the significant details including creator, funding, objective, stars system, schedule, telescope location, as well as data processing requirements. 6. The system provides generating status to astronomers.		
Subflows:		
Alternate/Exceptional Flow: 1. The user may not authorize at this level. 2. The astronomer does not fill in all the significant details.		

Activity diagram [Create a Science Plan]

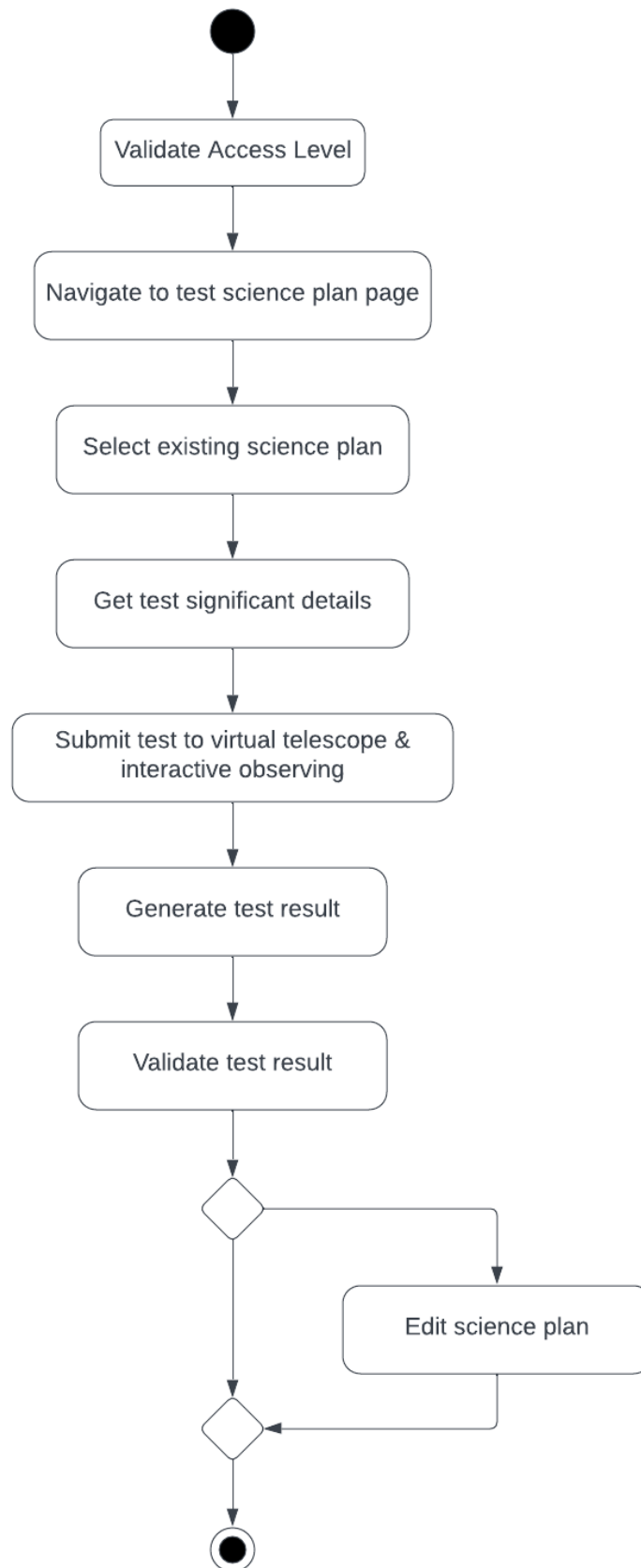


Sequence diagram [Create a Science Plan]

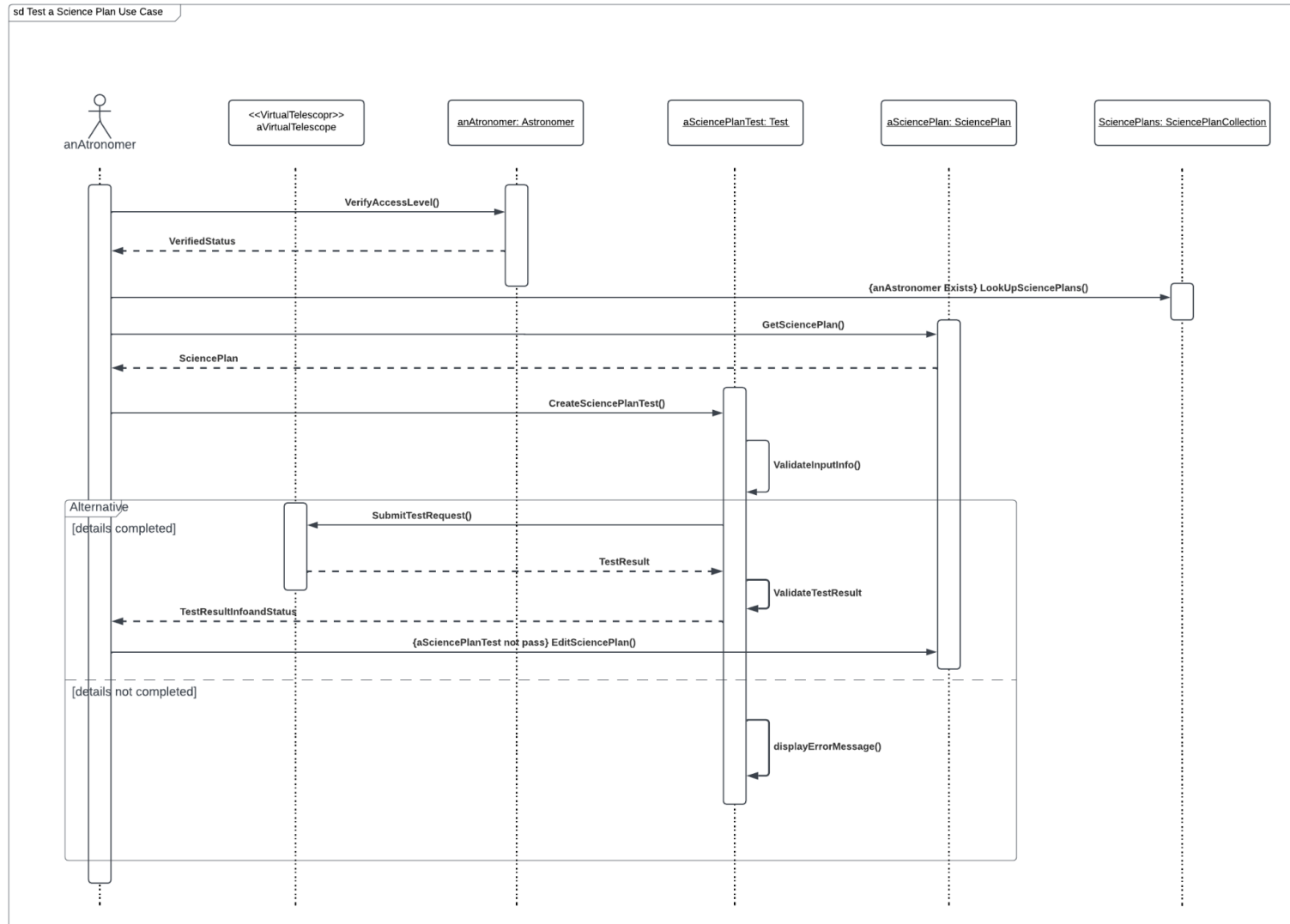


Use Case Name: Test a science plan	ID: 2	Importance Level: High
Primary Actor: Astronomer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science Observer		
Brief Description: This use case describes how a science plan is tested through the virtual telescope and interactive observing mode before submitting.		
Trigger: Astronomers need to validate the science plan whether it can operate on the telescope or not.		
Type: External		
Relationships: Association: Astronomer Include: Operate the interactive observing (Virtual Telescope) Extend: Generalization:		
Normal Flow of Events: 1. The system validates the user's level of access. 2. The system navigates the user to the science plan page. 3. The astronomer selects the existing science plan to test. 4. The system provides a test template that the astronomer can fill in. 5. The astronomer fills in with significant details of a test. 6. The system submits the test to the virtual telescope and interactive observing mode. 7. The system generates the test result and provides it to an astronomer. 8. The system validates the test result. - if the test does not pass the S-1: edit a science plan subflow is performed.		
Subflows: - S-1: Edit a science plan 1. The astronomer edits the current science plan. 2. The astronomer confirms the edited version of the science plan to the system.		
Alternate/Exceptional Flow: 1. The user may not authorize at this level. 2. The science plan that the astronomer requires does not exist yet. 3. The astronomer does not fill in all the significant details.		

Activity diagram [Test aScience Plan]

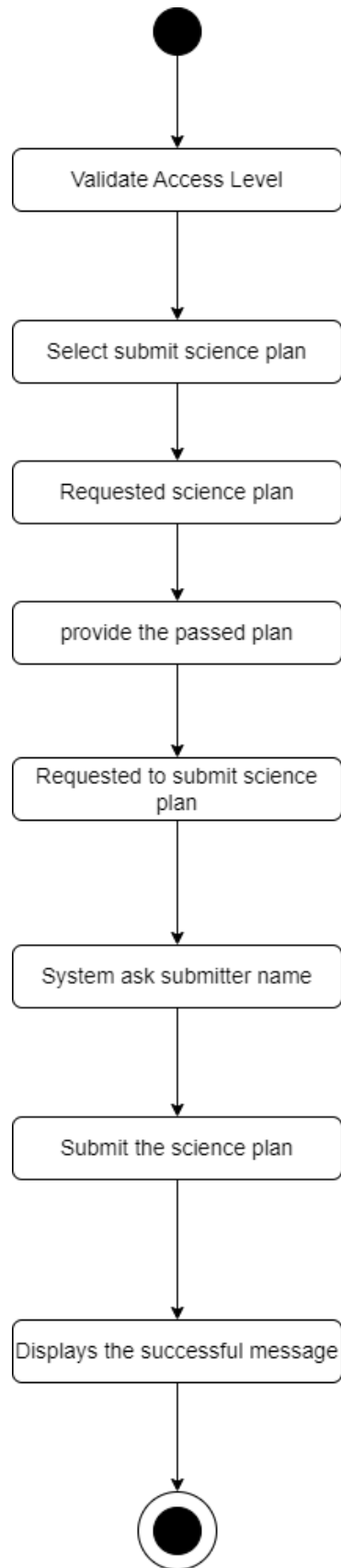


Sequence diagram [Test a Science Plan]

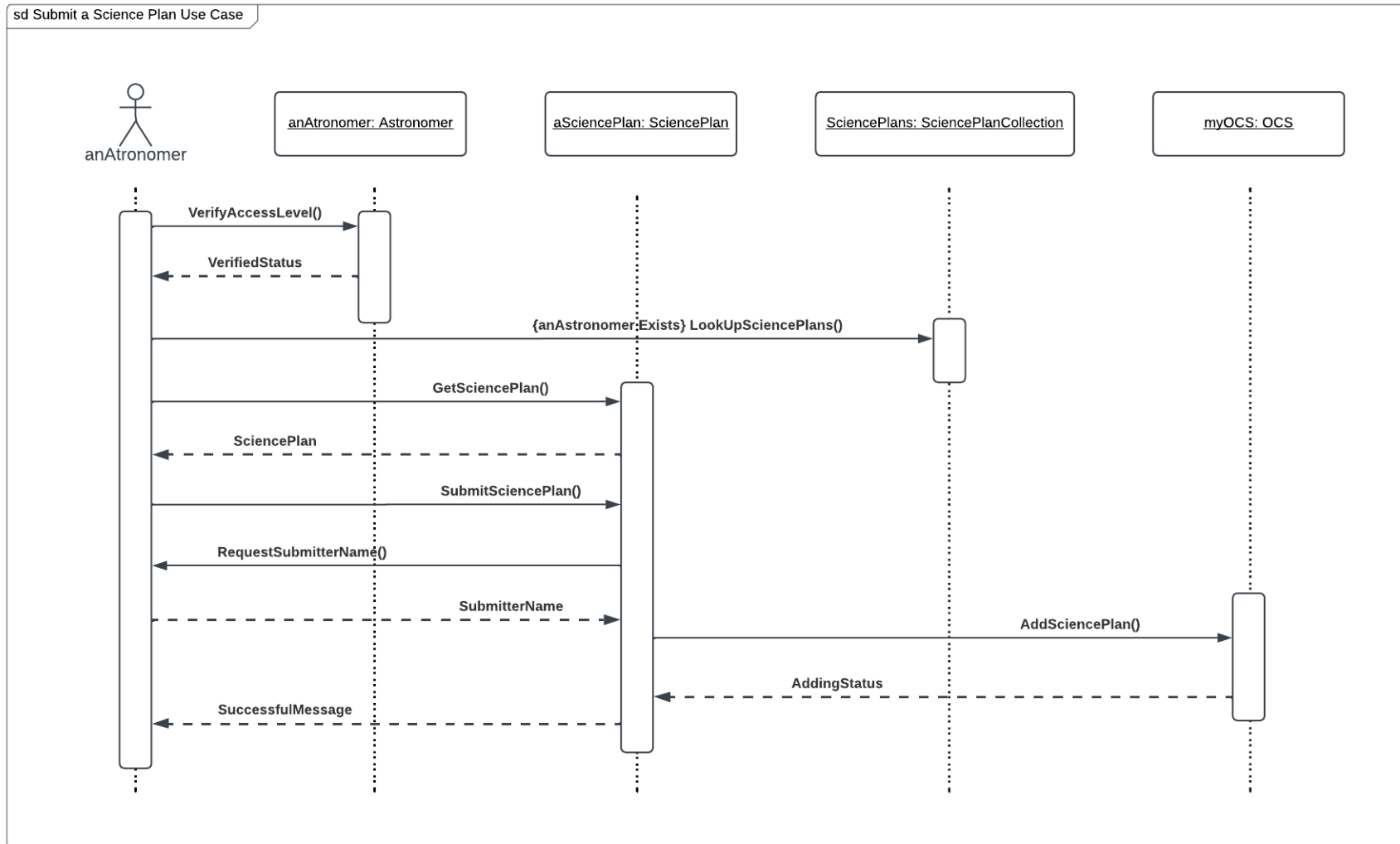


Use Case Name: Submit Science Plan	ID: 3	Importance Level: High
Primary Actor: Astronomer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science observer		
Brief Description: This use case describes how a science plan is submitted.		
Trigger: Astronomers need to submit a new science plan for collected astronomical data		
Type: External		
Relationships: Association: Astronomer Include: Extend: Generalization:		
Normal Flow of Events: 1. The system validates the user's level of access. 2. The astronomers select to submit a science plan. 3. The astronomers requested a science plan list by specifying the plan's name, plan status, and creator name. 4. The system provides the passed science plan. 5. The astronomer requests to submit the science plan to the system. 6. The system asks for his or her submitter's name. 7. The system displays the successful message back to the astronomers.		
Subflows:		
Alternate/Exceptional Flow: 1. The science plan that the astronomer requires does not exist yet. 2. The user requests a non-existing science plan.		

Activity diagram [Submit Science Plan]

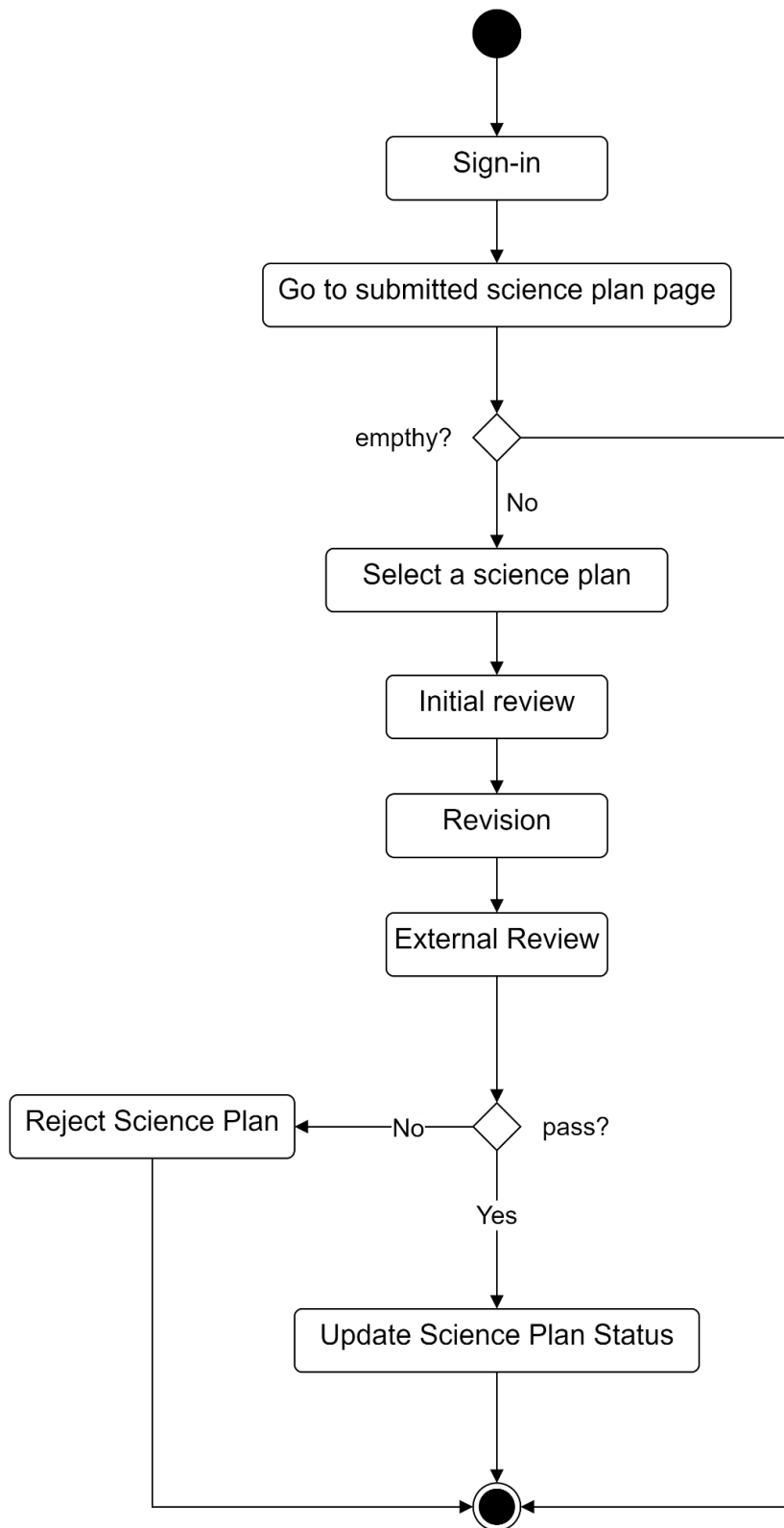


Sequence Diagrams [Submit Science Plan]

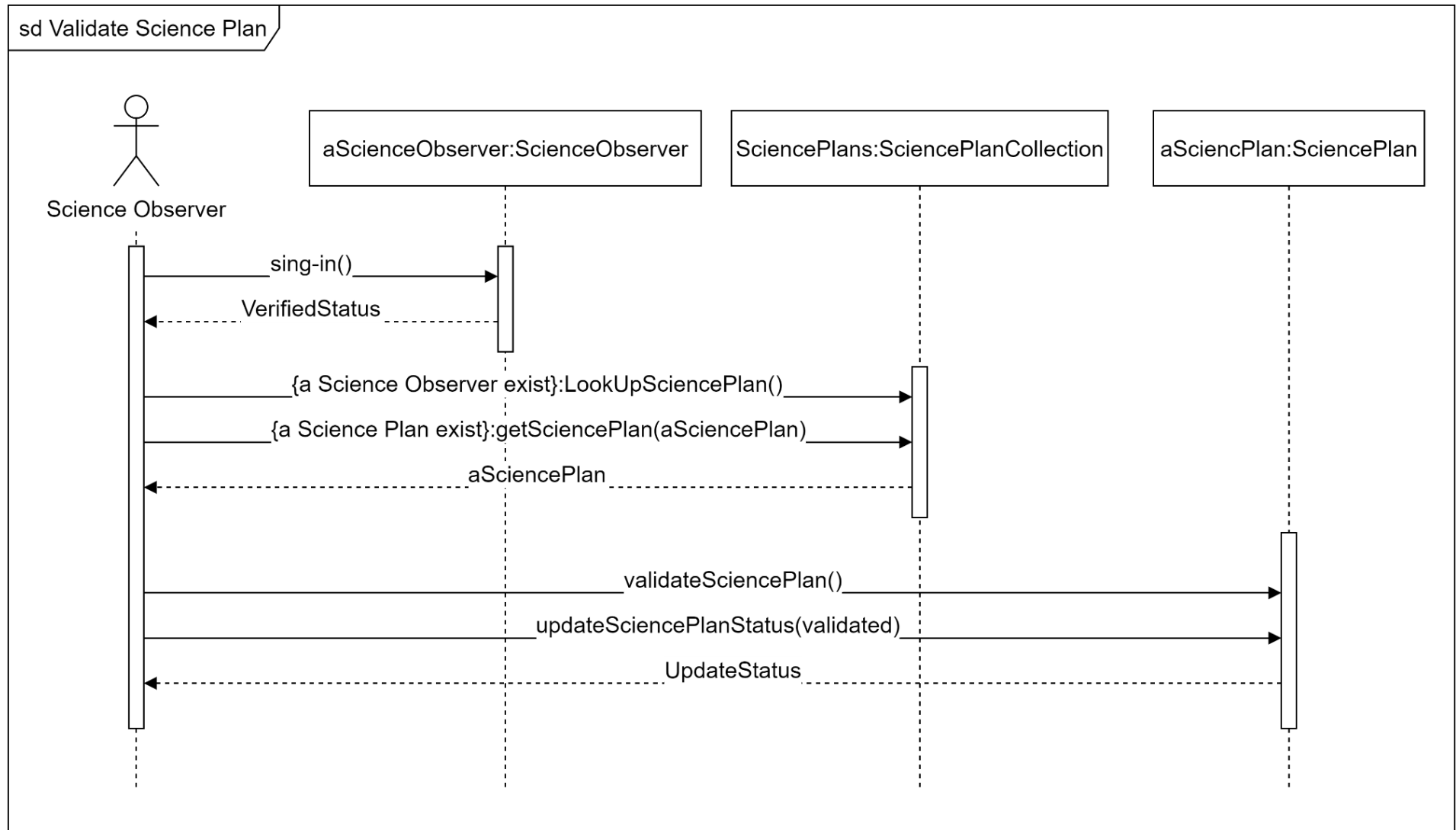


Use Case Name: Validate a science plan	ID: 4	Importance Level: High
Primary Actor: Science Observer		Use Case Type: Detail & Essential
Stakeholders and Interests: Science Observer - want to make sure the science plan work or not Astronomer - want to use their science plan to be the observing program		
Brief Description: This use case describe how science observer validate the science plan that created by an astronomer to further transform it to observing program.		
Trigger: When astronomer creates science plan Type: External		
Relationships: Association: Science Observer Include: Extend: Generalization:		
Normal Flow of Events: 1. The user(science observer) sign-in to the system. 2. The user goes to the submitted science plan page. 3. The user selects a science plan to be validated. 4. The user will do the initial review to ensure that it is scientifically sound and feasible, including schedule, telescope location, and data constrain. 5. 5. The user will do the revision based on the initial review feedback if any. 6. The broader user will do the external review of the revised plan, they may focus on the broader scientific context of the proposed program. 7. If the science plan passes all validation it will mark as validated else rejected. 8. The science plan was saved and then waiting to develop into observing program		
Subflows:		
Alternate/Exceptional Flow: 1. Submitted science plan is empty 2. System failure occurs		

Activity diagram [Validate science plan]

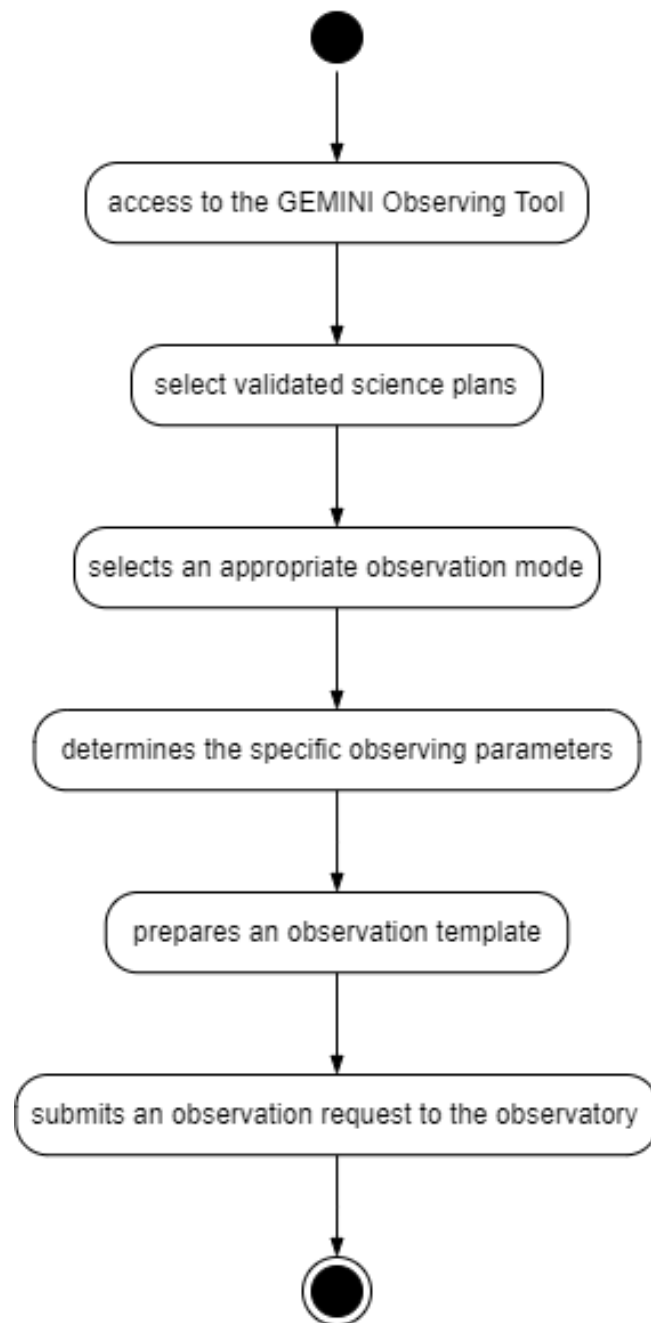


Sequence Diagram [Validate science plan]



Use Case Name: Create an observing program	ID: 5	Importance Level: high
Primary Actor: Science observer		Use Case Type: Detail & Essential
Stakeholders and Interests: Operation staff		
Brief Description: This use case describes how science observer creates an observing program from a science plan.		
Trigger: Science observer transform a science plan into an observing program		
Type: External		
Relationships: Association: Science observer Include: Extend: Generalization:		
Normal Flow of Events: 1. The user(science observer) sign-in to the system. 2. The scientific observer select validated science plans. 3. The scientific observer selects an appropriate observation mode, such as imaging or spectroscopy 4. The scientific observer determines the specific observing parameters required for the observation, such as exposure time, filter selection, and slit width 5. The scientific observer prepares an observation template that includes all of the necessary observing parameters by using the GEMINI Observing Tool. 6. The scientific observer submits an observation request to the observatory, specifying the desired observing dates and any other relevant information.		
Subflows:		
Alternate/Exceptional Flow:		

Activity diagram [Create an observing program]



Sequence Diagram [Create an observing program]

